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PDF Download
3616390.3618292.pdf
11 March 2026
Total Citations: 1
Total Downloads: 200

Published: 30 October 2023

Citation in BibTeX format

MSWiM '23: Int'l ACM Conference on Modeling Analysis and Simulation of Wireless and Mobile Systems
October 30 - November 3, 2023
Quebec, Montreal, Canada

Conference Sponsors:
SIGSIM

MobiWac '23: Proceedings of the Int'l ACM Symposium on Mobility Management and Wireless Access (October 2023)
https://doi.org/10.1145/3616390.3618292
ISBN: 9798400703676

Fast and Optimal Beam Alignment for Off-the-Shelf mmWave Devices

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ABSTRACT

Overcoming the very high path loss is considered a prerequisite for operating millimeter-wave (mmWave) beamforming to meet the high data-rate requirements of next-generation wireless systems. Existing work chose the optimal beam pattern outside the codebooks to maximize the SNR. However, it is nontrivial to quickly find the optimal beam for off-the-shelf mmWave devices, especially as the standardization efforts are moving towards incorporating large-scale antenna arrays. In this paper, we present *SWIFT*, a system that can significantly reduce the beam scan time while achieving optimal beam alignment for off-the-shelf mmWave devices. To implement *SWIFT*, we tackle algorithm complexity and calibration cost, making *SWIFT* independent of any phase measurements and instrument or specialized circuitry. Our results show that *SWIFT* can achieve near-optimal SNR performance on the off-the-shelf devices and significantly save beam scan time over the state-of-the-art. Our design is generic and has the potential to scale to future-large scale antenna arrays (e.g., massive MIMO in 5G and beyond).

CCS CONCEPTS

• **Networks → Wireless local area networks**.

KEYWORDS

millimeter wave, calibration, beamforming

ACM Reference Format:
Wei-Han Chen, Xin Liu, Kannan Srinivasan, and Srinivasan Parthasarathy. 2023. Fast and Optimal Beam Alignment for Off-the-Shelf mmWave Devices. In *Proceedings of the Int'l ACM Symposium on Mobility Management and Wireless Access (MobiWac '23), October 30-November 3 2023, Montreal, QC, Canada.* ACM, New York, NY, USA, 9 pages. https://doi.org/10.1145/3616390.3618292

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MobiWac '23, October 30-November 3 2023, Montreal, QC, Canada
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ACM ISBN 979-8-4007-0367-6/23/10…$15.00
https://doi.org/10.1145/3616390.3618292

1 INTRODUCTION

The large bandwidth available on mmWave carrier frequencies (e.g., 30-300 GHz) has spurred intense interest in exploring beamforming techniques in these frequency bands to meet the ever-increasing demands for high data rates. To establish a reliable mmWave communication link, previous research proposed various solutions [5, 21, 23, 25–27, 32] to search the pre-defined beam pattern in the codebooks of existing off-the-shelf devices, ensuring the transmitter and receiver beams are aligned. To further overcome the very high path loss in the frequency bands, the related work (*ACO* [20]) chose the optimal beam pattern outside the codebooks, which increases the SNR by a factor of 2.5 in the real-world environment.

However, it is nontrivial to quickly find the optimal beam for off-the-shelf mmWave devices, especially as the standardization efforts are moving towards incorporating large-scale antenna arrays for massive MIMO in 5G and beyond. This is because two issues need to be addressed: (1) **Algorithm Complexity.** As the phased array antenna size increases, the number of possible beam patterns grows exponentially, and it significantly raises the complexity of finding the optimal beam among all possibilities. A naive approach would be to search through all candidate beams exhaustively, but it does not scale well with the phased array size and will require considerable beam scan time ($O(4^N)$). While current state-of-the-art [20] can obtain the optimal beam, this approach still needs significant beam scan time ($O(N)$, detailed in Section 4). (2) **Calibration Cost.** Without calibration, the optimal beam deviates from its expected shape, i.e., leaks Radio Frequency (RF) energy into undesired directions. Therefore, the phased array antenna needs to be calibrated. However, the cost of calibrating hardware artifacts substantially increases [3, 4, 15, 31] as the phased array antenna size increase. Furthermore, the cost might increase because of the need for recalibration due to environmental changes, e.g., temperature drift.

In this paper, we present *SWIFT*, a system that can significantly reduce the algorithm complexity and calibration cost in order to achieve fast and optimal beam alignment for off-the-shelf mmWave devices. Moreover, *SWIFT* can achieve near-optimal throughput performance, reduce beam scan time significantly, and scale to future-large scale antenna arrays. To achieve our goal, however, the following challenges need to be overcome. Figure 1 visually summarizes the challenges.

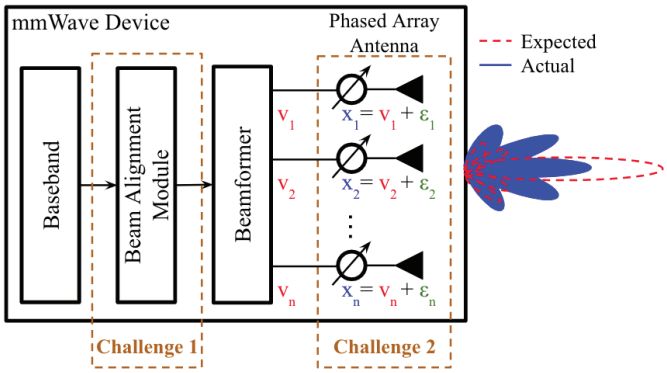

Figure 1: For large scale mmWave phased array antenna, it is challenging to find the optimal beam from a extremely large search space. Even such a beam is found $(v_1, v_2, ..., v_n)$, there are hardware artifacts whose ensembled effects are modeled as $(\epsilon_1, \epsilon_2, ..., \epsilon_n)$ creating a mismatch between the expected and actual beam.

C1: How to reduce the complexity of finding the optimal beam? As shown in the literature [28, 29], the mmWave channel is sparse, meaning that the number of propagation paths between a transmitter and a receiver is bounded by 3, and a dominant path usually exists. Leveraging this fact, we know that the optimal beam lies in a subspace corresponding to the underlying propagation paths, which is significantly smaller than the entire search space. The problem reduces to how to perform spatial sampling efficiently such that the physical parameters and the corresponding optimal beam can be estimated. Since we have reformulated the problem this way, compressive sensing might be a promising approach for our purpose. However, it is challenging to implement CS on off-the-shelf mmWave devices because it requires phase coherence across beam measurements that is unavailable on off-the-shelf mmWave radios [16, 22]. In our design, we perform spatial sampling and estimate the optimal beam from magnitude measurements only and do not rely on any phase measurements.

C2: How to reduce the cost of calibrating large-scale antenna arrays? For the calibration to be low-cost and widely applicable to all mmWave devices, including ones already on the market, our goal is to achieve calibration without using any instrument or specialized circuitry [3, 4, 15, 31]. To achieve this goal, we propose letting any two mmWave radios calibrate each other without instrument. By doing so, the calibration process can be performed anywhere and easily reinitiated whenever necessary. However, it is challenging to achieve this because of RF impairments such as Carrier Frequency Offset (CFO). When CFO is present, measurements made at different time points will have a different amount of random phase rotations. Thus, we can not measure phase information directly to calibrate against the hardware artifacts in the phased array antenna. In our design, we use Received Signal Strength (RSS) measurements to calibrate the hardware artifacts in the phased array antenna.

We implement our system, *SWIFT*, on an off-the-shelf mmWave device (detailed in Section 4.1) and demonstrates its effectiveness through real-world experiments. Our results show that *SWIFT* can achieve less than 1 dB median SNR loss compared to the optimal beam alignment and significantly save beam scan time over the state-of-the-art [20]. To summarize, this paper makes the following contributions:

- To the best of our knowledge, this is the first work that can achieve fast and optimal beam alignment on off-the-shelf mmWave devices, which significantly save beam scan time and maximize the SNR.
- We present the design of *SWIFT* that tackles the algorithm complexity and calibration cost, making our system independent of any phase measurements and instrument or specialized circuitry.
- We implement *SWIFT* on an off-the-shelf mmWave device and validate its effectiveness through real-world experiments. The results show *SWIFT* can achieve near-optimal SNR performance on the off-the-shelf devices and significantly save beam scan time over the state-of-the-art [20].

The rest of the paper is organized as follows. We introduce the background in Section 2. In Section 3, we present the system design of *SWIFT*. Evaluation results are reported in Section 4, and we describe the related work in Section 5 Finally, we conclude the paper in Section 6.

2 BACKGROUND

In this section, we introduce the mmWave channel model that is commonly used in the literature and establish the optimal beam for a given wireless channel. We assume a pair of mmWave transmitter (TX) and receiver (RX) are each equipped with a phased array antenna of size N_t and N_r, respectively. No assumptions are made on the geometry of the phased array antennas. To steer the beam, a TX can apply phase shifts to each antenna element to control the beam direction. Denote the signal sent from the TX as $\mathbf{x}$, and the received signal $\mathbf{y}$ at the RX can be expressed as:

$$\mathbf{y} = \mathbf{u}^{\mathbf{T}}\mathbf{H}\mathbf{v}\mathbf{x} + \mathbf{u}^{\mathbf{T}}\mathbf{n} \tag{1}$$

where $\mathbf{u}$ and $\mathbf{v}$ are the complex beamforming weights (phase shifts) applied at the RX and TX, respectively. The dimension of $\mathbf{u}$ and $\mathbf{v}$ are of $N_r \times 1$ and $N_t \times 1$, respectively. The Multiple-Input-Multiple-Output (MIMO) channel between the TX and RX is represented by the $N_r \times N_t$ matrix $\mathbf{H}$ and the vector $\mathbf{n}$ denotes the additive noise term with variance σ^2.

Next, we establish the optimal transmit beam for the TX for a given wireless channel $\mathbf{H}$. Without loss of generality, assume the RX is receiving with a beam $\mathbf{u}$, we define the receive channel as $\mathbf{H_{rx}} = \mathbf{u}^{\mathbf{T}}\mathbf{H}$. The received signal power and the noise power can be expressed as $|\mathbf{H_{rx}}\mathbf{v}|^2$ and $||\mathbf{u}||^2\sigma^2$, respectively. Since $||\mathbf{u}||$ is constant, maximizing SNR at the RX is equivalent to maximizing the received signal power. By Cauchy-Schwarz inequality [24], the optimal transmit beam at the TX can be found as the conjugate transpose of the receive channel $\mathbf{H}^*_{\mathbf{rx}}$. In Section 4, we will use $\mathbf{H_{rx}}$ to calculate the optimal beam.

As the phased array antenna size increases, the number of possible beams grows exponentially. It is nontrivial to quickly find the optimal beam from such a large search space. Next, we will explain

how to achieve fast and optimal beam alignment for large-scale mmWave antennas.

3 *SWIFT* DESIGN

In this section, we detail how to achieve fast and optimal beam alignment for large scale mmWave phased array antenna. Our design is composed of two parts: (1) a beam alignment module that reduces the complexity of finding the optimal beam (Section 3.1) and (2) a calibration module that lowers the cost of removing hardware artifacts from the phased array antenna (Section 3.2) . *SWIFT* works by calibrating the mmWave radio once and in the communication stage the beam alignment module enables fast and optimal beam alignment.

3.1 Reducing Algorithm Complexity

Our insight is to leverage the sparsity of the mmWave channel such that the optimal beam can be estimated through spatial sampling, which considerably reduces the search space for the optimal solution. Key differentiators of our approach from extant solutions [16, 22] is that our approach (1) does not assume phase coherence across beam measurements that is not available on off-the-shelf mmWave radios and (2) estimates the optimal beam from magnitude measurements only, which makes it widely applicable to off-the-shelf mmWave radios.

Next, we explain how to perform spatial sampling efficiently and estimate the optimal beam from the physical parameters. For simplicity, we assume the TX uses an Uniform Linear Array (ULA) phased array antenna of size N_t. For a propagation path with a AoD of θ, the corresponding *spatial frequency* for this θ can be written as:

$$w = \frac{2\pi d cos(\theta)}{\lambda} \tag{2}$$

where d and λ are the antenna spacing and the wavelength of the central frequency, respectively. We note that spatial frequency is a function of both array geometry and AoD and Equation 2 will change if the TX is not using a ULA. For the AoD θ, the corresponding *array response* for each transmit antenna element can be written as a column vector:

$$A(w) = [1, e^{jw}, e^{j2w}, ..., e^{j(N_t-1)w}]^T \tag{3}$$

Assume the TX sends a total of M measuring probes and for the i-th probe the beamforming weight is set as $p_i = [1, e^{j\phi}, e^{j\phi_2}, ..., e^{j\phi_N}]^T$, where $[\phi, \ldots, \phi_N]$ are independent and identically distributed random variables from a uniform distribution on $(0, 2\pi)$. To efficiently sample the propagation path in space, we desire the probes to be as spatially uncorrelated as possible. Consider there exists a dominant single path with spatial frequency w_0 between the TX and RX and the corresponding complex channel gain is h. The i-th Received Signal Strength (RSS) measurement is expressed as:

$$y_i = |hA(w)^T p_i + n_i| \tag{4}$$

where n_i denotes white additive noise with variance σ^2. Let the nominal RSS pattern for spatial frequency w be denoted as $x(w) = |A(w)^T p|$ and the spatial frequency of the propagation path can be estimated though maximizing the cost function:

$$\hat{w} = \arg\max_{w} \langle \frac{y}{||y||}, \frac{x(w)}{||x(w)||} \rangle \tag{5}$$

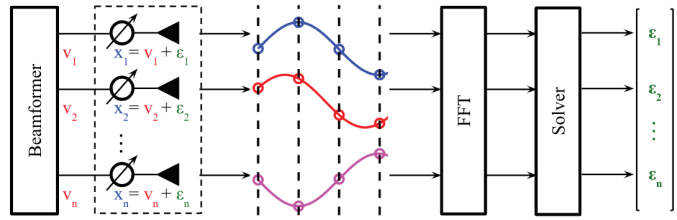

Figure 2: An overview of how the calibration is performed in *SWIFT*. For each antenna element, the collected power measurements form a sinusoid, which is then analyzed by FFT. A solver is used to eliminate the path effects from the ensemble and finally the phase offsets $(\epsilon_1, \epsilon_2, ..., \epsilon_n)$ are derived.

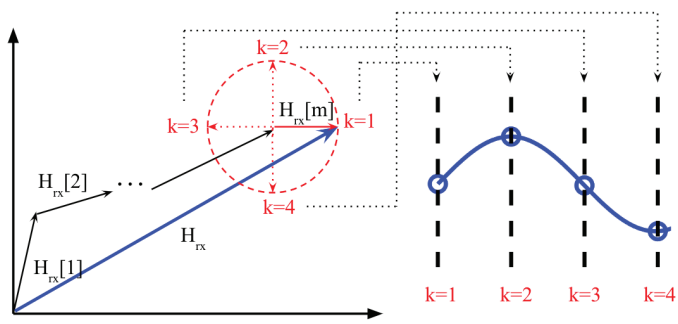

Figure 3: An example of measuring the phase of the receive channel for the m-th antenna element. For the k-th measuring probe, a phase shift of $(k-1)\pi/2$ is applied on the m-th antenna while the phase shifts for the rest of the antenna elements remain unchanged.

Further optimization can be performed to refine the estimation results. This concludes how we perform spatial sampling for 1D array. The model can be easily extended to accommodate 2D array by introducing two spatial frequencies for azimuth and elevation angles. From Equation 4 and Equation 5, since our design deals with real numbers instead of complex numbers, *SWIFT*'s beam alignment complexity is no more than the complexity of compressive sensing $O(K \log N_t)$ [16, 22], where K denotes number of existing propagation paths.

3.2 Lowering Calibration Cost

The mismatch between the expected and actual beam pattern must be removed to achieve optimal beamforming. The reason for the mismatch is that there exists hardware artifacts whose ensembled effects are modeled as phase offsets between the antenna elements.

Next, we will explain how to obtain phases of the receive channel $\mathbf{H_{rx}}$ from RSS measurements (Section 3.2.1) and use them to calibrate against the hardware artifacts (Section 3.2.2). Figure 2 gives an overview of the calibration process in *SWIFT*.

3.2.1 Phase Retrieval. In this section, the goal is to measure the phase of the receive channel $\mathbf{H_{rx}}$ (see Section 2) from the RSS measurements. Without loss of generality, we can assume that the TX first find a beamforming vector $\mathbf{v'}$ such that it can reach the

RX with a decent SNR and the RX uses a fixed receive beam. We refer to this transmit beam $\mathbf{v}'$ as the reference and the phase offsets are computed relative to the reference. We define the phase of the reference beam as zero, i.e., $\angle \mathbf{H_{rx}}\mathbf{v}' = 0$. To find out the relative phase offset of m-th antenna element to the reference, i.e., $\angle \mathbf{H_{rx}}[m]$, the TX sends probes with the m-th antenna element changing phase shifts successively while the rest antenna elements remain using the same phase shift as the reference beam. Our experiment platform supports 2-bit phase shift registers, that is, for an antenna element we can apply phase shift values of $[0, \pi/2, \pi, 3\pi/2]$. Thus, for the m-th antenna element, it will iterate through four changing phase shifts. Specifically, for the k-th measuring probe, the TX uses the beam vector $\mathbf{v_k}$ that is defined as:

$$\mathbf{v_k}[i] = \begin{cases} e^{j(k-1)\frac{\pi}{2}}, & \text{if } i = m \\ \mathbf{v}'[i], & \text{otherwise} \end{cases} \tag{6}$$

The resultant k-th power measurement $x_k = |\mathbf{H_{rx}}\mathbf{v_k}|^2$ can be written as:

$$\begin{aligned} x_k &= |(\mathbf{H_{rx}}\mathbf{v}' - \mathbf{H_{rx}}[m]\mathbf{v}'[m]) + \mathbf{H_{rx}}[m]e^{j(k-1)\frac{\pi}{2}}|^2 \\ &= |r_1 e^{j\theta_1} + r_2 e^{j\Delta_k}|^2 \\ &= |(r_1 cos(\theta_1) + r_2 cos(\Delta_k)) + j(r_1 sin(\theta_1) + r_2 sin(\Delta_k))|^2 \\ &= (r_1^2 cos^2(\theta_1) + r_2^2 cos^2(\Delta_k) + 2r_1 r_2 cos(\theta_1) cos(\Delta k)) + \\ &(r_1^2 sin^2(\theta_1) + r_2^2 sin^2(\Delta_k) + 2r_1 r_2 sin(\theta_1) sin(\Delta k)) \\ &= r_1^2 + r_2^2 + 2r_1 r_2 cos(\Delta_k - \theta_1) \\ &= r_1^2 + r_2^2 + r_1 r_2 e^{j(\theta_2-\theta_1)} e^{j(k-1)\frac{\pi}{2}} + r_1 r_2 e^{-j(\theta_2-\theta_1)} e^{-j(k-1)\frac{\pi}{2}} \end{aligned} \tag{7}$$

where $r_1 = |\mathbf{H_{rx}}\mathbf{v}' - \mathbf{H_{rx}}[m]\mathbf{v}'[m]|$, $\theta_1 = \angle(\mathbf{H_{rx}}\mathbf{v}' - \mathbf{H_{rx}}[m]\mathbf{v}'[m])$, $r_2 = |\mathbf{H_{rx}}[m]|$, $\theta_2 = \angle \mathbf{H_{rx}}[m]$, and $\Delta_k = \theta_2 + (k-1)\frac{\pi}{2}$. Equation 7 shows that the RSS measurements $|\mathbf{H_{rx}}\mathbf{v_k}|^2$ form a sinusoid and thus can be efficiently analyzed by a standard Fast Fourier Transform (FFT). Figure 3 visualizes the process of changing phase shifts for measuring the phase offset of the m-th antenna element. Next, to determine the phase of the sinusoid relative to the reference, we construct a vector from the four RSS measurements as $\mathbf{x} = [x_1, x_2, x_3, x_4]$. From the Inverse Discrete Fourier Transform expression [19], we have the Fourier Transform of $\mathbf{x}$ as the coefficients of the complex exponentials:

$$\mathcal{X} = \mathscr{F}(\mathbf{x}) = [r_1^2 + r_2^2, r_1 r_2 e^{j(\theta_2-\theta_1)}, 0, r_1 r_2 e^{-j(\theta_2-\theta_1)}] \tag{8}$$

The individual quantity r_1 and r_2 are unknown but can be expressed in terms of $\mathcal{X}$. Consider the following two binomial expansions:

$$\begin{aligned} (r_1 + r_2)^2 &= r_1^2 + r_2^2 + 2r_1 r_2 = \mathcal{X}[1] + 2|\mathcal{X}[2]| \\ (r_1 - r_2)^2 &= r_1^2 + r_2^2 - 2r_1 r_2 = \mathcal{X}[1] - 2|\mathcal{X}[2]| \end{aligned} \tag{9}$$

Given the expansions, r_1 and r_2 can be reorganized as:

$$\begin{aligned} r_1 &= \begin{cases} \frac{\sqrt{\mathcal{X}[1]+2|\mathcal{X}[2]|}+\sqrt{\mathcal{X}[1]-2|\mathcal{X}[2]|}}{2}, & \text{if } r_1 \geq r_2 \\ \frac{\sqrt{\mathcal{X}[1]+2|\mathcal{X}[2]|}-\sqrt{\mathcal{X}[1]-2|\mathcal{X}[2]|}}{2}, & \text{otherwise} \end{cases} \\ r_2 &= \begin{cases} \frac{\sqrt{\mathcal{X}[1]+2|\mathcal{X}[2]|}-\sqrt{\mathcal{X}[1]-2|\mathcal{X}[2]|}}{2}, & \text{if } r_1 \geq r_2 \\ \frac{\sqrt{\mathcal{X}[1]+2|\mathcal{X}[2]|}+\sqrt{\mathcal{X}[1]-2|\mathcal{X}[2]|}}{2}, & \text{otherwise} \end{cases} \end{aligned} \tag{10}$$

From Equation 10, there will be two sets of outcome for r_1 and r_2 and the ambiguity can be eliminated by occurrence of a contradiction. Ultimately, our goal is to determine $\angle \mathbf{H_{rx}}[m]$. Given $\angle \mathcal{X}[2] = \theta_2 - \theta_1$, we can write $\angle \mathbf{H_{rx}}[m] = \theta_2 = \angle \mathcal{X}[2] + \theta_1$. Thus, next we want to express θ_1 in terms of $\mathcal{X}$. By definition, the reference beam has zero phase, i.e., $\angle(\mathbf{H_{rx}}\mathbf{v}') = 0$ and we have the following derivation for θ_1:

$$\begin{aligned} (\mathbf{H_{rx}}\mathbf{v}' - \mathbf{H_{rx}}[m]\mathbf{v}'[m]) + \mathbf{H_{rx}}[m]\mathbf{v}'[m] &= \mathbf{H_{rx}}\mathbf{v}' \\ r_1 e^{j\theta_1} + r_2 e^{j\theta_2}\mathbf{v}'[m] &= \mathbf{H_{rx}}\mathbf{v}' \\ r_1 + r_2 e^{j(\theta_2-\theta_1)}\mathbf{v}'[m] &= \mathbf{H_{rx}}\mathbf{v}' e^{-j\theta_1} \\ \angle(r_1 + r_2 e^{j(\angle\mathcal{X}[2])}\mathbf{v}'[m]) &= -\theta_1 \end{aligned} \tag{11}$$

Finally, following the result of Equation 11, we can now write $\angle \mathbf{H_{rx}}[m]$ in terms of $\mathcal{X}$:

$$\angle \mathbf{H_{rx}}[m] = \angle \mathcal{X}[2] - \angle(r_1 + r_2 e^{j(\angle\mathcal{X}[2])}\mathbf{v}'[m]) \tag{12}$$

where r_1 and r_2 are defined in Equation 10. The process can be repeated to obtain the phases for all antenna elements on the TX, i.e., the full $\angle \mathbf{H_{rx}}$.

Next, we will explain the number of measurements required to compute $\mathbf{H_{rx}}$ for a N-element phased array antenna. For measuring the m-th antenna element, there are two cases: the m-th antenna was on in the reference beam ($|\mathbf{v}'[m]| > 0$) and the m-th antenna was off in the reference beam ($|\mathbf{v}'[m]| = 0$). If the m-th antenna was on in the reference beam, one of the four phase shifts for the m-th antenna has been measured in the reference beam and three additional probes are required. On the other hand, if the m-th antenna was off in the reference beam, it requires four additional probes. Therefore, the total number of measurements required is bounded between $1 + 3N$ and $1 + 4N$.

3.2.2 Phase Offset Correction. In this section, we will explain how to infer the phase offsets between the antenna elements from the obtained $\mathbf{H_{rx}}$. The receive channel $\mathbf{H_{rx}}$ is a function of both the receive beam $\mathbf{u}$ and the wireless channel $\mathbf{H}$. While the receive beam $\mathbf{u}$ can be fixed such that it does not introduce different phase responses for different TX antenna elements, the wireless channel $\mathbf{H}$ creates different phase responses on the antenna elements based on the path angles AoA and AoD. Specifically, the exact responses are dictated by the mapping functions $S_r(\cdot)$ and $S_t(\cdot)$ that map AoA and AoD to their corresponding array responses, respectively. If there are array responses from the wireless channel mixed with the phase offsets between antenna elements, we can not separate the desired phase offsets from the ensemble. To understand why, let's revisit Equation 1 and introduce the phase offsets between antenna elements at the TX as a $N_t \times N_t$ diagonal matrix $E = diag([e^{j\epsilon_1}, ..., e^{j\epsilon_{N_t}}])$:

$$\mathbf{y} = \mathbf{u}^{\mathbf{T}}\mathbf{H}(\mathbf{E}\mathbf{v})\mathbf{x} + \mathbf{u}^{\mathbf{T}}\mathbf{n} \tag{13}$$

Thus, the measured $\mathbf{H_{rx}}$ now includes the phase offset terms

$$\begin{aligned} \mathbf{H_{rx}} &= \mathbf{u}^{\mathbf{T}}\mathbf{H}\mathbf{E} \\ \mathbf{H_{rx}}[\mathbf{k}] &= \sum_{m=1}^{N_r} u_m h_{mk} e^{j\epsilon_k} \end{aligned} \tag{14}$$

where

$$\mathbf{H} = \begin{bmatrix} h_{11} & h_{12} & \dots & h_{1N_t} \\ h_{21} & h_{22} & \dots & h_{2N_t} \\ \vdots & \vdots & \vdots & \vdots \\ h_{N_r1} & h_{N_r2} & \dots & h_{N_rN_t} \end{bmatrix}, \mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ n_{N_r} \end{bmatrix}.$$

From Equation 14 we can see that there are 2 unknowns per $\mathbf{H_{rx}}[k]$, i.e., $\sum_m u_m h_{mk}$ and $e^{j\epsilon_k}$. For the full $\mathbf{H_{rx}}$, there are N_t equations and $2N_t$ unknowns, which forms an under-determined system of linear equations that has infinitely many solutions. Therefore, the phase offsets can not be uniquely determined from the ensemble in general.

However, under certain circumstances, obtaining the phase offsets from $\mathbf{H_{rx}}$ is possible. One such case is when there exists a single dominant path between the TX and RX and the AoD is known. First, the phase offsets are relative quantities and the exact values are not of our interest, i.e., knowing the phase differences between $\mathbf{H_{rx}}[k]$ suffices. Thus, to figure out the phase offset on the k-th TX antenna ϵ_k from $\mathbf{H_{rx}}$, we do not need to know the receive beam $\mathbf{u}$ or the wireless channel $\mathbf{H}$ exactly. Second, only the AoD is of importance and the AoA is irrelavant because the wireless channel is comprised of a single dominant path. Consider the RX uses a fixed receive beam $\mathbf{u}$, the receive channel for the k-th TX antenna can be rewritten as:

$$\begin{aligned} \mathbf{H_{rx}}[k] &= \sum_{m=1}^{N_r} u_m h_{mk} e^{j\epsilon_k} \\ &= \sum_{m=1}^{N_r} u_m [S_r(AoA)_m S_t(AoD)_k] e^{j\epsilon_k} \\ &= e^{j\beta} S_t(AoD)_k e^{j\epsilon_k} \end{aligned} \tag{15}$$

where β is a real number and remains the same for all k. This shows that the AoA only introduces a common scaling factor for the receive channel but does not change the relative quantity. A change in AoA simply results in a different common scaling factor $e^{j\beta}$. If the TX has the knowledge of $S_t(\cdot)$ and the path angle AoD, then there is exactly one unknown per $\mathbf{H_{rx}}[k]$ since the contribution from $S_t(AoD)_k$ can be eliminated.

Given the above discussion, we propose to put the TX and RX in LOS with zero azimuth and elevation angles to perform calibration. This makes the calibration setup simple and obviates the need of measuring the AoD. Additionally, our proposed method works on phased array antennas that have arbitrary antenna placements in 3D space as well. As long as the mapping function $S_t(\cdot)$ is known, the geometry of the phased array antenna does not impose limitations on the viability of the calibration.

To summarize, *SWIFT* achieves low-cost calibration for mmWave phased array antenna without using any equipment or specialized circuitry [3, 4, 15, 31]. The only requirement is the knowledge of the mapping function $S_t(\cdot)$, which is not a problem for device vendors since they have the phased array antenna design schematics. Importantly, *SWIFT* can be easily implemented on off-the-shelf mmWave radios through software updates and does not require any changes in hardware.

4 EVALUATION

In this section, we detail our implementation, experiment setup, and performance evaluation. We also compare the performance of our proposed work with the state-of-the-art beam alignment algorithms.

4.1 Implementation

We implemented our system on an off-the-shelf mmWave radio [34], as shown in Figure 4. We use a USRP radio as the Baseband Processor Unit (BPU) and signal processing is done by using MATLAB. Our mmWave radio has eight 6×6 rectangular phased array antenna modules. The antenna modules have 2-bit phase shifters for coarse phase shift control. Each antenna module takes as input 15 GHz IF data signals and up-convert them to 60 GHz for mmWave communications. To generate user-defined 15 GHz IF data signals, a customized circuit board is used to mix the IF output signals from the USRP with the clock signal from the on-device Qualcomm QCA6335 baseband chip. For real-time beam steering, the antenna modules are controlled by a customized FPGA that sends switching commands at specified timestamps. Beam patterns are defined by the codebook entries which specify the indices for the gain control registers and the phase shift registers. The codebook can be modified by loading customized entries into the device's internal memory through Ethernet connection. To achieve correct beamforming, the phased array antenna control commands and the IF data signals need to be synchronized. We achieve the synchronization by using the Automatic Transmit/Receive (ATR) function supported by the USRP. The FPGA is triggered to send control commands to the phased array antenna by the GPIO pins on the USRP when it enters transmit state.

4.2 Experiment Setup

Experiments are done in a 24×13 meters office with multiple reflectors, e.g., metal cabinets, whiteboards, and concrete walls. We randomly place the TX and RX at various locations in the office to cover indoor LOS and NLOS scenarios typical for IEEE 802.11ad communication. At each location, the RX receives with a wide beam and the TX iterates through different beam alignment algorithms. In total, we collected more than 700 data traces. The calibration is performed once by placing the TX and RX 3 meters apart in direct LOS.

4.3 Compared Schemes

Beam alignment algorithms can be divided into two categories: codebook-based and codebook independent methods. Codebook-based methods (e.g., [1, 21, 25, 33]) try to select the best entry from the predefined codebook for communication. While utilizing a codebook reduces implementation complexity and generally achieves good coverage, it does not guarantee optimal performance. For a given environment, the performance is bounded by the codebook if the optimal beam is not within the codebook. On the other hand, codebook-independent methods (e.g., [5, 20, 26, 27]), generally gather information from the wireless channel and create the best beam accordingly. We include 802.11ad [1] and ACO [20] as the representative algorithms for each category for our performance

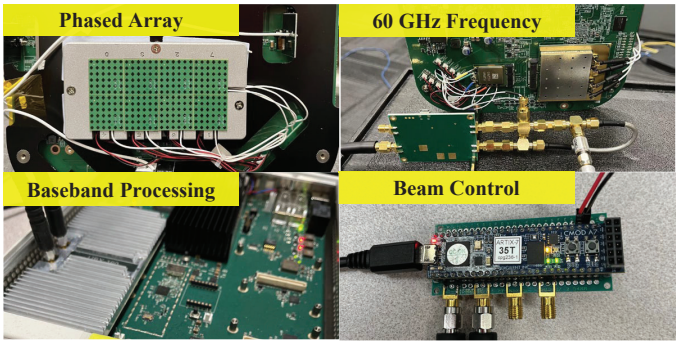

Figure 4: Our mmWave radio platform with eight 6×6 rectangular phased array antenna modules.

comparison. We list the algorithms for our performance comparison as follows:

- **802.11ad** [1]. We include 802.11ad because it serves as a representative algorithm for codebook-based methods and is widely used for 60 GHz WLAN. We use the default codebook with 128 entries configured by the vendor. The best beam is determined by probing through the entire codebook.
- ***ACO*** [20]. *ACO* is the state-of-the-art beam alignment algorithm that can achieve optimal beamforming. Specifically, ACO is able to extract the full CSI without phase-coherent measurements and calculate the optimal beam for any given wireless channel. We note that ACO can achieve optimal beamforming even there are hardware artifacts in the phased array antenna. The phase offsets will manifest as part of the phase responses in the CSI and therefore will be accounted for when calculating the optimal beam.
- **Exhaustive search**. This brute-force method can find the optimal beam by exhaustively searching through all possibilities. The number of beam probes scales exponentially with the phased array antenna size and the resolution of the phase-shifters.
- ***SWIFT***. Our proposed method effectively removes the mismatch between expected and actual beam patterns and can achieve near-optimal performance for large phased array antennas while significantly reduces beam scan time.

4.4 Micro-benchmark

To verify the calibration works, we investigate the SNR loss of exhaustive search compared to the optimal before and after calibration in LOS conditions. If the generated beam pattern follows the expected beam shape, the RF energy is concentrated within the mainlobe and results in high SNR when aligned with the LOS path. Since ACO is resilient to hardware artifacts in the phased array antenna, we use ACO to find the optimal beam. Figure 5 shows the CDF. Due to phase offsets between antenna elements, the generated beam from the uncalibrated phased array antenna has more RF energy leaked into undesired directions, i.e., stronger sidelobes, and thus it has degraded SNR when aligned with the LOS path. After calibration, the generated beam is closer to the one calculated by the theoretical model and results in better beam alignment. The median SNR loss are 0.11 and 2.15 dB for calibrated and uncalibrated exhaustive search, respectively. Even at 90th percentile, calibrated exhaustive search has less than 1 dB SNR loss while the uncalibrated exhaustive search has over 4 dB SNR loss. This shows that the calibration is effective.

4.5 Overall Performance

In this section, we compare the performance of each scheme against the optimal. Each scheme collects 128 measurements before they make a beamforming decision and the optimal beam is found by using *ACO*. Figure 6 shows the CDF of SNR loss of each scheme compared to the optimal. The median SNR loss are 0.17, 3.20 ,and 2.79 dB for *SWIFT*, 802.11ad, and exhaustive search, respectively. First, *SWIFT* achieves near-optimal performance with a median SNR loss of only 0.17 dB. Even at 90th percentile point, we observe that it has less than 1 dB SNR loss in our experiments. Second, we see that the exhaustive search can not achieve optimal beamforming. The exhaustive search supposedly can find the optimal beam by probing through all possible beams. However, due to the mismatch between expected and actual beams, the optimal beam deviates from its beam shape and leaks RF energy into undesired directions. As a result, the signal energy from the TX does not propagate through to the RX as expected. Third, 802.11ad also does not achieve optimal beamforming. Since 802.11ad finds the best beam by probing through all the beams in the codebook, it does not guarantee to find the optimal beam for a given wireless environment.

Next, we investigate the throughput performance of each scheme. Figure 7 shows the CDF of the data rate performance. The median data rate are 3.85, 3.85, 2.5, and 2.5 Gbps for *ACO*, *SWIFT*, 802.11ad, and exhaustive search, respectively. We observe that the throughput performance of *SWIFT* closely follows the optimal and has a median throughput gain of 1.54x over 802.11ad and exhaustive search. The hardware artifacts in the phased array antenna can reduce the median throughput by more than 1 Gbps for the exhaustive search.

4.6 Impact of Number of Measurements

In this section, we investigate how the number of measurements affects the performance of each scheme, excluding exhaustive search. For codebook-based beam selection methods, scanning more beams increases the chance of achieving better alignment. For *ACO*, the number of required probes depends on the phased array antenna size and it might need to probe up to $1+4N$ beams, where N denotes the phased array size. When the number of probes is less than the required, *ACO* can not fully utilize all antenna elements and has to beamform with a subset of the antennas. In other words, the performance of *ACO* increases as the number of probes increases until it reaches the maximum number of required probes. On the other hand, *SWIFT* spatially samples the optimal signal direction and can find the optimal beam for a given wireless channel with logarithmic number of measurements. *SWIFT* can fully utilize all antenna elements for beamforming once the optimal signal direction is identified. The beam selection for *SWIFT* becomes more stable as the number of measurements increases.

Figure 8 shows the individual performance of SNR loss compared to the optimal as the number of measurements increases.

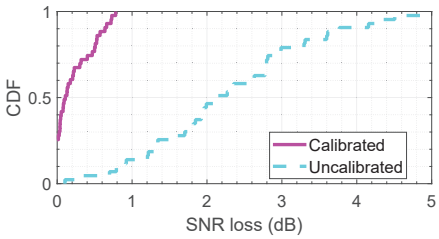

Figure 5: Micro-benchmark of our mmWave phased array antenna calibration module. Our calibration improves the median SNR performance by 2.15 dB for exhaustive search.

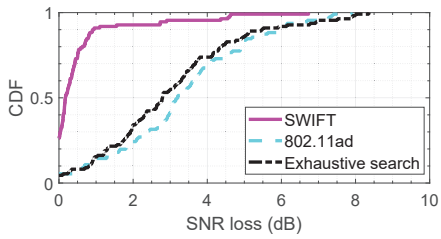

Figure 6: *SWIFT* achieves a median SNR loss of 0.17 dB compared to the optimal and has a median of 3.03 and 2.62 dB SNR gain over 802.11ad and exhaustive search, respectively.

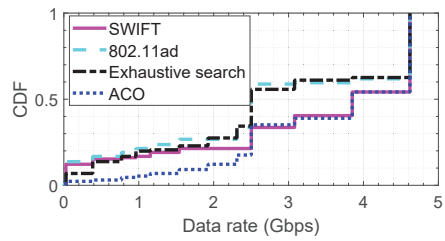

Figure 7: *SWIFT* achieves near-optimal throughput performance and has a median throughput gain of 1.35 Gbps over 802.11ad and exhaustive search.

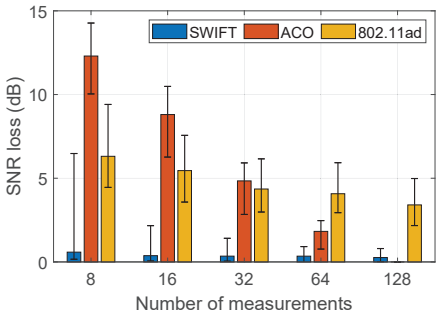

Figure 8: *SWIFT* requires as little as 8 measurements to achieve a median SNR loss of less than 0.5 dB compared to the optimal while *ACO* needs over 64 measurements to achieve such SNR performance. The performance of 802.11ad improves as it scans more beams but it still has a large gap from the optimal.

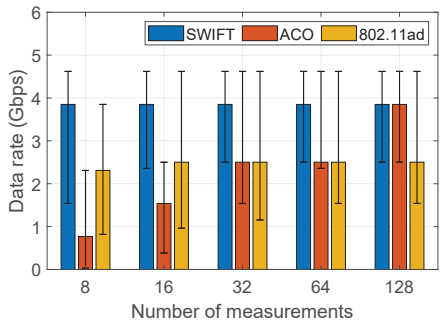

Figure 9: *SWIFT* can achieve a median throughput of 3.85 Gbps with only 8 measurements while *ACO* requires 128 measurements to achieve comparable throughput performance. *SWIFT* has throughput gain over the other two of at least 1.54x and up to 5x given the same number of measurements used.

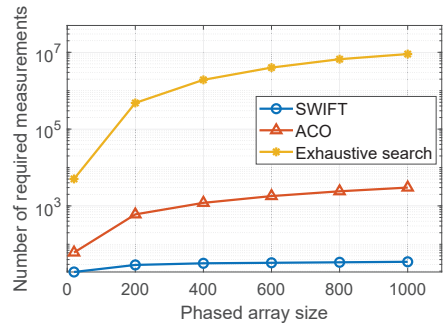

Figure 10: *SWIFT*, *ACO*, and exhaustive search scale logarithmically, linearly, and exponentially with the phased array antenna size, respectively. We plot the lower bound for exhaustive search, which scales quadratically with the phased array antenna size when a dominant LOS path exists.

First, *SWIFT* performs very well and requires as little as 8 measurements to achieve a median SNR loss of less than 0.5 dB. In contrast, using the same number of measurements, *ACO* has over 12 dB SNR loss. Second, *ACO*'s SNR loss reduces as the number of activated antenna elements increases. Last, when the number measurements increases from 8 to 128, the median SNR loss has a 3 dB improvement for 802.11ad. Although probing more beams increases performance for 802.11ad, we observe that it is still far from the optimal. Figure 9 shows the throughput performance for each scheme as the number of measurements increases. First, we observe that the median throughput gain of *SWIFT* over the other schemes is at least 1.54x and up to 5x given the same number of measurements used. To achieve the optimal throughput performance, *SWIFT* significantly reduces the number of required measurements compared to *ACO*. Evidently, this indicates that *SWIFT* can achieve fast and optimal beam alignment on large phased array antennas. The impact of reduced number of measurements on beam scan time is discussed next in Section 4.7.

4.7 Scalability

In this section, we investigate the scalability of *SWIFT* with other two schemes that can achieve optimal beam alignment, i.e., *ACO* and exhaustive search. Beam alignment algorithm complexity determines the number of required measurements for large scale mmWave phased array antennas. First, *SWIFT* can find the optimal alignment in $O(K \log N)$ measurements, where K and N denote the number of propagation paths that exist between the TX and RX and the number of antennas, respectively. Previous studies have shown that K is bounded by 3 [28, 29]. Second, the number of required measurements for *ACO* is between $1 + 3N$ and $1 + 4N$ as discussed in Section 3.2.2. Last, exhaustive search finds the optimal beam in $O(4^N)$ measurements, assuming the phase shifters have 2-bit phase control. Even if we are considering LOS scenarios only, exhaustive search still requires $O(N^2)$ measurements. The number of required measurements grows prohibitively large for *ACO* and exhaustive search as the phased array antenna size increases while *SWIFT* scales well with the phased array size. We show the number of required measurements as a function of phased array antenna size in Figure 10. Here, we conservatively set $K = 3$ and assume

ACO requires minimum amount of measurements $1 + 3N$. Thus, the figure shows the minimum number of measurements *SWIFT* can save over the other two schemes.

The beam scan time for a client is a function of the number of measurements that need to be collected and the collision probability, which depends on MAC layer mechanisms. For example, in 802.11ad, each client needs to contend for beam training slots within each beacon interval that typically has a period of 100 ms [1]. The collision probability depends on the number of contending clients under the same network. There are up to 128 beam training slots in one beacon interval for 802.11ad. If a device can not successfully send all its probes without collision, it has to wait for another beacon interval to continue to contend for beam training. Conservatively speaking, the beam scan time reduces at least 1.516 seconds per 1024 beams saved for 802.11ad assuming zero collision probability. As the collision probability increases, the beam scan time for *SWIFT* can only improve since it requires significantly less number of measurements than ACO and exhaustive search.

5 RELATED WORK

mmWave beam alignment. Many mmWave beam alignment algorithms have been proposed over the past few years. Some works [5, 7, 18, 23, 26, 32] propose to incorporate out-of-band information to assist beam selection. These works require extra sensors at the TX, such as light sensor [5], camera [7, 32], MIMO WiFi radios [18, 26], or LiDAR and GPS [23]. Different from these methods, *SWIFT* identifies the optimal beam by performing spatial sampling with in-band mmWave signals only and does not require extra non-RF sensors. Some machine learning approaches [21, 33] aim to train a deep neural network model that can quickly select the best beam from the codebook. However, their beam selection is limited by the codebook whereas *SWIFT* can find the optimal beam beyond what a codebook can offer. Recent compressive sensing solutions [16, 22] and AgileLink [6] enable a TX to achieve beam alignment with logarithmic number of measurements. However, they either require phase coherence across measuring probes [16, 22] or a customized phased array antenna [6] that are not available on commodity mmWave radios. Different from these approaches, *SWIFT* can work with low-cost commodity mmWave radios and does not require phase coherence across measurements. The hierarchical beam search proposed in IEEE 802.11ad standard [1] can take up to few seconds to find the correct beam [35]. BeamSpy [29] infers the best beam with fixed overhead but it is limited to quasi-stationary links only. Different from these methods, our work allows for fast beam alignment in typical indoor environments and can adapt quickly to dynamic channels. UbiG [27] solves for parameters of each physical path through a genetic algorithm and assembles the optimal beam from the estimated parameters. However, UbiG's algorithmic complexity incurs significant computational overhead, which limits its deployability on commodity mmWave devices. In contrast, *SWIFT* can be applied to commodity mmWave devices to obtain the optimal beam.

ACO [20] is perhaps the most relevant work to ours. *ACO* aims to extract the full CSI from carefully crafted probes and calculates the optimal beam. However, the number of required probes becomes prohibitively large as the phased array antenna size increases. In contrast, our work requires significantly less number of measurements that scale logarithmically with the phased array antenna size.

Phased array antenna calibration. There are many techniques proposed to perform phased array antenna calibration. A series of work proposes to deploy a Built-In-Self-Test (BIST) circuitry [3, 4, 8–10, 15] to achieve on-device self-calibration. Rotating element electric field vector-based (REV) methods use amplitude measurements to calculate the phase difference in the phased array antenna [2, 12–14, 30]. In [17], the authors proposed to calibrate the phased array antenna through online channel measurements during communication. However, this method can not be realized in practice since phase-coherence is not available on off-the-shelf mmWave radios. Similar to REV methods, *SWIFT* calibrates off-the-shelf mmWave radios by power measurements only, obviating any phase measurements, instrument, or specialized circuitry.

6 CONCLUSION

In this paper, we present *SWIFT*, a system that can significantly reduce the algorithm complexity and calibration cost in order to achieve fast and optimal beam alignment for off-the-shelf mmWave devices. To implement *SWIFT*, we tackle algorithm complexity and calibration cost, making *SWIFT* independent of any phase measurements or specialized circuitry. Our experiment results show that *SWIFT* can achieve near-optimal SNR performance on the off-the-shelf devices and significantly save beam scan time over the state-of-the-art (*ACO* [20]). Our design is generic and has the potential to scale to future-large scale antenna arrays.

ACKNOWLEDGMENTS

This work was supported in part by NSF Award 2112471 and Cisco Award PO USA000EP279201.

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